INSECTICIDAL EFFECTS OF CAPSICUM OFFSPRING EXTRACT ON PERIPLANETA AMERICANA (COCKROACH)

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APPROVAL SHEET

IN PARTIAL FULFILLMENT of the requirements for the Academic Track, SCIENCE, TECHNOLOGY, ENGINEERING, AND MATHEMATICS, this research entitled "**THE EFFECT OF INTERNET ACCESS RELIABILITY TO STUDENTS TIMELINESS ON ONLINE TASK SUBMISSION**", prepared and submitted by Group 4 In Practical Research 2 has been examined, accepted, and approved for PROPOSAL HEARING.

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DEDICATION

This research study is wholeheartedly dedicated to all students who aspire to explore science beyond the classroom. May this work inspire you to conduct meaningful research, ask critical questions, and seek innovative solutions to real-world problems through knowledge and determination.

We dedicate this study to future researchers who wish to expand the field of botanical insecticides and sustainable pest management. May this research serve as a foundation for further scientific investigations involving Capsicum extracts and their insecticidal properties against Periplaneta americana and other household pests. We hope that future studies will refine extraction methods, improve concentration optimization, and contribute to safer and more environmentally friendly alternatives to synthetic insecticides.

This study is also dedicated to homeowners and local communities who seek affordable, accessible, and safer pest control solutions. May the findings of this research encourage the use of plant-based alternatives that minimize chemical exposure while maintaining effectiveness.

To environmental advocates and health-conscious individuals, we dedicate this work as a contribution toward promoting sustainable practices in urban pest management. May this research support the movement toward reducing harmful chemical pollutants in households and communities.

We also dedicate this research to our families, whose sacrifices, encouragement, and unwavering support became our motivation to persevere throughout the challenges of experimentation and data gathering.

Above all, this study is dedicated to God Almighty, for granting us wisdom, patience, and strength in completing this academic endeavor.

ABSTRACT

American cockroaches (Periplaneta americana) still persist as domestic pests that can cause the transmission of diseases and spoilage of food. In addition, the continued use of synthetic insecticides poses risks to the environment and human health. The present study was conducted to explore the safer and environmentally friendly properties of using Capsicum offspring as insecticide agents. The objectives of the research were to determine the mortality rate of Periplaneta americana exposed to different concentrations (100%, 75%, 50%, 25%) of Capsicum annuum and Capsicum frutescens extracts as natural insecticides. Furthermore, it aimed to determine whether there is a significant difference in the mortality rates of Periplaneta americana exposed to different concentrations of Capsicum annuum and Capsicum frutescens compared to the insecticide. A quantitative research design was used in this study. In this design, Periplaneta americana specimens were directly exposed to different concentrations of Capsicum annuum and Capsicum frutescens. The results showed that both Capsicum annuum and Capsicum frutescens achieved 100% mortality at 100% concentration. Capsicum annuum achieved complete mortality in exactly 6 minutes and 46 seconds, whereas Capsicum frutescens achieved complete mortality at exactly 15 minutes. At 75%, Capsicum annuum achieved partial mortality, whereas Capsicum frutescens achieved complete mortality. At 50%, Capsicum annuum predominantly caused temporary paralysis, whereas Capsicum frutescens achieved 80% mortality. At 25%, both Capsicum annuum and Capsicum frutescens functioned as irritants. The results showed that there is a significant difference in the mortality rates of Periplaneta americana exposed to different concentrations of Capsicum annuum and Capsicum frutescens compared to the insecticide ($F(8,81)=9.6118 < 0.05$). The results of the study showed that Capsicum offspring have high potential as environmentally safe insecticides.

Keywords: *American cockroach (Periplaneta americana) , Capsicum "offspring" extract, botanical insecticide, capsaicinoids, mortality rate, sustainable pest management.*

CHAPTER I

THE PROBLEM AND ITS SCOPE INTRODUCTION

Rationale of the Study

Cockroaches are among the most persistent household pests worldwide and are commonly associated with poor sanitation and the transmission of disease-causing microorganisms. The American cockroach (Periplaneta americana) is particularly problematic due to its high adaptability to human environments, rapid reproductive capacity, and increasing resistance to many commercially available chemical insecticides. These characteristics make infestations difficult to control and pose significant risks to public health. Cockroaches are known carriers of various pathogens that can contaminate food and surfaces, especially in residential areas, food establishments, and healthcare facilities. As a result, their presence is not only a nuisance but also a serious concern in maintaining hygiene and preventing disease transmission. These challenges highlight the need for more effective, safe, and sustainable pest control alternatives.

The plants are naturally synthesized secondary metabolites like chemical compounds but not directly includes growth but important for defense against herbivores animals and different pests. In terms of species, the belonging of genus Capsicum, capsaicinoids such as capsaicin and dihydrocapsaicin are the primary secondary metabolites that is responsible for pungency and have been massively involved with deterrent and toxic effects on insects and other arthropods. These different types of compounds are new to Capsicum and it works as a natural defense chemical that can lower the insects feeding, survival, and infestation. In addition, to capsaicinoids, Capsicum plants have other bioactive compounds, that includes flavonoids, alkaloids, and phenolic compounds, which significantly improves their insect-deterrent properties and support their potential use as plant-based insecticides. (Batiha et. al, 2020)

According to Alarcon et al., a plant-based insecticides have attracted interest because of their natural origins and reduced ecological impact. The Capsicum genus, which includes chili peppers such as Capsicum annuum and Capsicum frutescens contains capsaicinoids bioactive compounds known for their pungency and defensive role against insects. Studies have shown that chili pepper extracts exhibit toxic effects on insects by disrupting their nervous system and physiological functions. A laboratory study demonstrated that Capsicum annuum extract caused

significant mortality in cockroaches, with higher concentrations resulting in increased death rates, indicating its potential as a natural insecticidal agent. (Alarcon et al., 2024).

Another evidence supports the insecticidal properties of Capsicum species beyond cockroaches. Research on capsaicinoids extracted from Capsicum Frutescens revealed strong insecticidal activity against agricultural pests such as aphids, with mortality increasing as extract concentration increased (showing high percentages of aphid mortality up to 99% under laboratory conditions and up to 97% in field applications). These findings suggest that capsaicinoids possess broad insecticidal potential and can affect different insect species through similar toxic mechanisms (Claros Cuadrado et al., 2019). Such results strengthen that the chilibased extracts may be effective against pests like cockroaches.

Some empirical studies gives strong evidences that can support the insecticidal activity of Capsicum extracts, specifically against cockroach pests. Methanolic and ethyl acetate extracts of Capsicum annuum fruits have shown significant impact as insecticidal effects against Periplaneta americana, a species that is closely related to Blatta orientalis. Controlled bioassays demonstrates that methanolic extracts achieved 100% mortality at 80% concentration within 24 hours, while ethyl acetate extracts can cause lower mortality, manifested specifically through sluggish movement and partial paralysis. These findings shows that extraction solvent and concentration play an important role in identifying whether the chili based insecticidals are effective. The mortality that was observed to capsaicin and other bioactive fatty acids, such as oleic and palmitic acids, identified through GC-MS analysis. (Roheem and Hamisu, 2025)

However, existing literature predominantly focuses on agricultural pests such as aphids and beetles, often emphasizing repellent effects rather than lethal dose comparisons or focusing strictly on single-species Capsicum evaluation. The reliance shows that solvent-based extraction systems for general pest control rather than species-specific evaluation of Periplaneta americana.

Consequently, there is limited empirical research directly comparing multiple Capsicum species, specifically annuum, frutescens, and chinense, in a single controlled experimental design against Periplaneta americana, particularly when using offspring-derived extracts and evaluating their efficacy against commercial insecticides. Additionally, while botanical insecticides are often promoted as resistance-mitigating alternatives Isman (2006), a very few controlled experimental

studies to quantify mortality differences among Capsicum species varieties under the standardized exposure conditions using statistical comparison, such as ANOVA.

Thus, the present study addresses this research gap by examining three Capsicum variants and statistically analyzing mortality differences, thereby contributing a novel, species-specific, and comparative data to the growing field of botanical urban pest management.

Not only adult cockroaches, Capsicum annuum extracts has potential insecticidal effect that has also been documented against larval stages of medically important insect pests. Furthermore, the observed toxicological showed complete mortality (100 MO%) in Culex pipiens mosquito larvae from petroleum ether extracts of chili peppers that provided and significant toxicity in Musca domestica larvae after 24 hours of exposure, with defined LC_{50} values. These results suggest that the complex phytochemical composition of Capsicum extracts comprising capsaicinoids, flavonoids, and phenolic acids may act synergistically to disrupt insect physiological and metabolic processes across multiple life stages. (Baz et. al 2025)

Theoretical Background

The rising need for environmentally friendly pest control methods which sustain natural ecosystems has created increased interest in botanical insecticides as effective solutions that replace synthetic chemical pesticides. The use of traditional pesticides proves effective yet these products create environmental damage which poses health threats to humans and enables pests to build resistance against them. The accessibility of plant-derived extracts which biodegrade in nature and show low toxicity to mammals has become important to developing countries and tropical regions. The theoretical foundation of this study rests on three scientific and environmental theories which together explain how plant secondary metabolites interact with Periplaneta americana through biochemical and physiological and ecological pathways.

The Neurotoxic Interference Theory serves as the main foundation of this study which states that plants produce specific secondary metabolites which include capsicum capsaicinoids as their natural defense system to disrupt predator nervous system functions. In Periplaneta americana, capsaicin and related compounds interfere with neural signaling pathways by activating ion channels and sensory neurons associated with pain and stress responses. The biochemical overload

results in three possible outcomes which include paralysis and metabolic exhaustion and death. The research team determines the extract concentration level to be the most important factor which controls how many subjects will die. The study by Hayman and Kam (2010) demonstrates that capsaicinoids produce severe physiological stress in arthropods which indicates that higher concentrations of offspring seed-heavy extracts will create stronger insecticidal effects because they contain more alkaloids.

The Cuticular Disruption Theory (Ebeling, 1971) supports anchor theory by showing that insect survival depends on the complete integrity of their exoskeletons. The theory proposes that specific botanical oils and compounds function as surfactants or abrasive agents which dismantle the waxy lipid barrier found on the cockroach cuticle. The Oriental cockroach Periplaneta americana displays higher susceptibility to desiccation because its epicuticle contains both more porous material and greasy substances than other cockroach species when it comes into contact with acidic and oily elements from Capsicum extract. The study conducted by Nyamador et al. (2017) established that botanical extracts kill insects through two different methods: ingestion and contact toxicity which leads to rapid death because the extract prevents insects from maintaining their moisture balance.

The Secondary Metabolite Synergy Theory (Berenbaum, 1985) shows that plant extract mixtures create stronger effects than single chemical components. Capsicum species contain their main toxic agent capsaicin along with saponins and flavonoids and essential oils. The two compounds function together because saponins enable cockroaches to absorb neurotoxic capsaicin through their body membranes. The research conducted by Isman (2006) discovered that botanical insecticides achieve better results because complex plant mixtures create obstacles for pests to develop resistance through multiple chemical attack methods.

In the Philippines indigenous materials serve as effective pest control solutions because of the native presence of Capsicum annuum (Siling Labuyo) and the expensive nature of synthetic pesticides. Agbayani (2021) tested local chili extracts against common household pests and discovered that capsaicin solutions functioned as strong repellents yet the research found that "offspring" seed extracts got thrown away instead of being used. This research examines how to develop "offspring" extract into an insecticidal product that will kill insects instead of just repelling them.

The research conducted by Mizutani et al. in (2025) investigates the neurophysiological theory which states that capsaicin performs its insecticidal function through its ability to block voltage-gated sodium channels (VGSCs) based on different states. The research conducted by Mizutani et al. in (2025) demonstrates that capsaicin does not block these channels through random methods because it specifically targets sodium currents when the channels reach their slow-inactivated state. The compound establishes the inactive status of the nerve cell through its ability to create a negative shift on the steady-state inactivation curve which prevents the central nervous system from producing spontaneous activity and causes the insect to experience fast paralysis. Mizutani et al. (2025) conducted their research which appears in Toxicon to demonstrate how capsaicin acts as a natural neurotoxicant to kill American cockroaches and other species.

The researchers Kamezaki and Nishiwaki (2025) investigate whether capsaicin functions as an insect neurotoxicant by attacking voltage-gated sodium channels despite pyrethroid resistance. According to Kamezaki and Nishiwaki (2025) capsaicin exhibits a distinct statedependent inhibitory mechanism which remains effective against knockdown-resistant (kdr) strains of German cockroaches (Blattella germanica) that contain the L993F mutation. The study shows that this mutation leads to a major reduction in traditional pyrethroid sensitivity because capsaicin can still block sodium currents and cause paralysis through a different binding site and inhibitory pathway from standard insecticides. Kamezaki and Nishiwaki (2025) demonstrate that capsaicin functions as a biochemical method which effectively controls insect populations that have developed resistance to pesticides through its ability to preferentially block channels which exist in their slow-inactivated state.

The research conducted by Maliszewska et al. (2018) establishes a theoretical framework which demonstrates that invertebrates develop heat sensitivity through particular pharmacological compounds. The study demonstrates that American cockroaches (Periplaneta americana) develop desensitization through repeated exposure to specific alkaloids. The study demonstrates that capsaicin and capsazepine and camphor cause cockroaches to respond to noxious temperatures (50°C) with increased escape latency and shift their behavioral thermopreference toward warmer environments. The study shows that cockroaches use a TRPmediated thermosensory system which functions like the system that mammals use because repeated agonist application leads to "whole-animal desensitization" which prevents the insect from protecting itself against overheating. The

study shows that these compounds block heat nociception but they do not affect how insects respond to noxious cold or how their essential physiological markers like catalase activity change, which shows a specific neurotoxic effect rather than general systemic failure.

The theoretical background of this study highlights that Capsicum-derived extracts have strong potential as an effective botanical insecticide against Periplaneta americana. The discussion demonstrates how capsaicin and other secondary metabolites create multiple pathways which disrupt nerve signaling and damage the insect cuticle and enhance toxicity through chemical synergy to produce paralysis desiccation and death according to Neurotoxic Interference Theory and Cuticular Disruption Theory and Secondary Metabolite Synergy Theory. The empirical studies demonstrate that capsaicin affects specific neurophysiological mechanisms which include voltage-gated sodium channels in pesticide-resistant cockroach strains. The targeted action demonstrates its effectiveness while synthetic pesticides create a higher risk. The research needs to be conducted because the Philippines has native Capsicum annuum plants and people have not yet discovered all ways to use their seed-based extracts. The theoretical framework demonstrates that Capsicum offspring extracts can serve as sustainable and accessible and environmentally safe methods for controlling cockroaches.

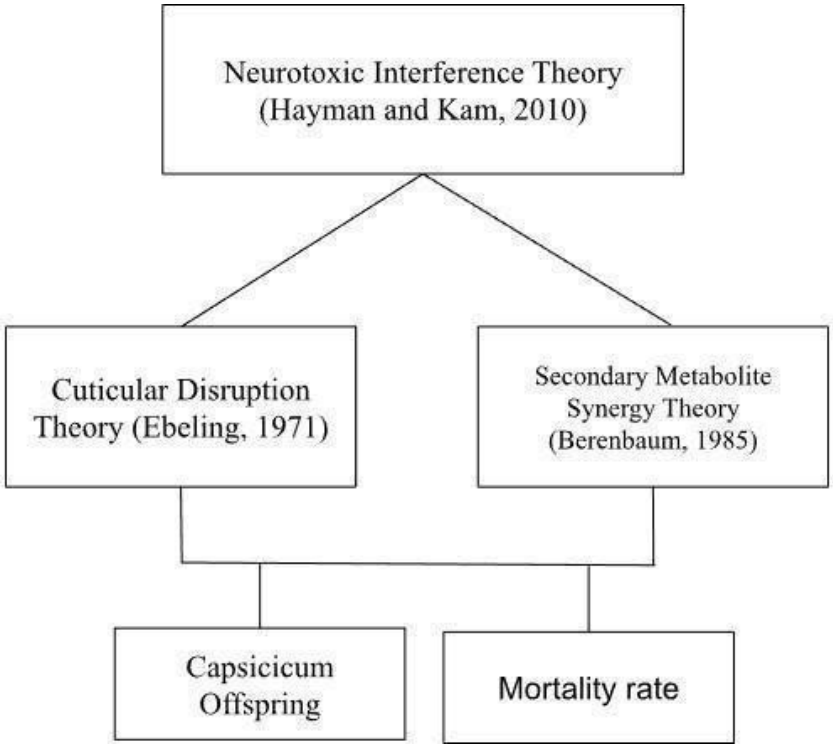

Figure 1. Theoretical Conceptual Framework

THE PROBLEM

Statement of the Problem

This study is aiming to determine the insecticidal effects of Capsicum Offspring extract on Periplaneta americana (cockroach) under ordinal experiment conditioning.

Specifically, it seeks to answer the ff questions:

1. What is the mortality rate of Periplaneta americana treated with chili extract compared to those treated with commercial insecticide?
2. What is the mortality rate of Periplaneta americana when exposed to Chili (Capsicum extract) in concentration in the ff variants?

    2.1 Capsicum Annuum;

    2.2 Capsicum Frutescens?

3. Is there a significant difference on mortality rate within group and across groups?

Hypothesis of the Study

H_0: There is no significant difference in the mortality rates across all experimental and control groups.

H_1: There is a significant difference in the mortality rates across the different groups, indicating that the type and concentration of extract impact the survival of the cockroaches.

Scope and Delimitation

The scope of this research will be centered on assessing the insecticidal qualities of the extracts made from three particular varieties of the Capsicum species, to wit: Capsicum annuum (Bell/Chili pepper) and Capsicum frutescens (Siling Labuyo). The main aim of this research will be to quantify the rate of mortality among the American cockroach (Periplaneta americana) within a predetermined observation period. The research will also involve a comparative study between the organic extracts and a chosen commercial insecticide to ascertain whether natural alternatives can be as effective as synthetic chemical pesticides. The research will utilize statistical analysis to determine whether there are any differences in the potency between the various pepper varieties.

The study of this research will be limited to the use of the Periplaneta americana species and will not cover other varieties of cockroaches or common household pests. The strength of the extracts will be held constant based on the researcher's "ordinal experiment conditioning," and the research will not delve into the particular molecular composition of capsaicinoids or the long term effects of the extract on the environment. Moreover, the research will be performed in a controlled laboratory or home setting, which means that the results of the experiment may not be generalizable to open-air agricultural settings.

Significance of the Study

For the homeowner and the community, this study represents a safer, more natural alternative to chemical pesticides, providing a means of controlling Periplaneta americana infestations without the use of toxic chemicals. For health and environmental professionals, the results of this study offer information on effective pest control practices that are safe and minimize the chemical runoff and respiratory problems associated with traditional pesticides. Product developers and entrepreneurs may also find the results of this study useful, as it illustrates the potential for developing cheap and biodegradable insecticides from local botanical sources such as Capsicum. For the researcher, this study represents an increased understanding of botanical toxicology and the application of scientific principles, assisting in both academic and individual growth. Finally, future researchers can build on the results of this study and examine the effectiveness of different varieties of chili peppers or test the extract against other common

household pests, working towards a more sustainable and environmentally friendly method of urban pest control.

DEFINITION OF TERMS

This section is an important part of the study as it explains the main concepts and variables being examined. By giving clear and exact definitions, it helps maintain consistency and clarity in how terms are understood throughout the research process.

Capsicum Annuum. A chili pepper plant species. In this experiment, it is one of the three types of chilies used to create the "offspring extract" to be tested for its insect-killing properties.

Capsicum Frutescens. It is a wild or cultivated chili pepper, sometimes including Siling Labuyo. It is used as one of the treatments given to Blatta orientalis in this experiment.

Chili Offspring Extract. This refers to the liquid from the fruits or seeds of the secondgeneration (offspring) chili plants. This is the independent variable to be tested for its ability to kill or repel cockroaches.

Commercial Insecticide. This is a mass-produced chemical pest killer bought from the market. This is used as the Positive Control group to compare the natural chili extract to the standard chemical solutions.

Insecticidal Effect. This refers to the ability of a substance to kill insects. In this experiment, it is determined by the mortality rate (the number of cockroaches killed) within a given time frame after exposure.

Mortality Rate. This is the number of deaths in a group. In practical terms, it is determined by dividing the number of dead Periplaneta americana by the total number of subjects in each group after treatment.

CHAPTER II

REVIEW OF RELATED LITERATURE AND RELATED STUDIES

This section presents the literature review and relevant studies that underpin the research inquiries of the investigation. It focuses on the insecticidal properties and applications of Capsicum annuum and Capsicum frutescens, highlighting both international and local studies.

Capsicum Annuum

In recent years, there has been growing interest in the use of plant-based insecticides as alternatives to synthetic chemical pesticides. Botanical insecticides are generally considered safer, more environmentally friendly, and less likely to contribute to insect resistance. Among these, Capsicum annuum, commonly known as chili pepper, has gained attention due to its insecticidal properties. This plant contains bioactive compounds, particularly capsaicinoids such as capsaicin, which have been reported to exhibit toxic, repellent, and growth-inhibiting effects on various insect species. Several studies have explored the potential of C. annuum extracts in controlling household pests, including cockroaches (Periplaneta americana). Isman (2006) emphasized that botanical insecticides derived from plants like chili peppers affect the insect nervous system and feeding behavior. Similarly, Tavares et al. (2018) found that higher concentrations of chili pepper extracts significantly increased mortality rates in insects such as cockroaches and ants. These findings support the potential use of C. annuum as an effective and sustainable alternative to conventional chemical insecticides.

A study published in the Journal of Entomology (2016) evaluated the toxicity and repellency of various plant extracts, including C. annuum, when applied individually and in combination against termites (Macrotermes bellicosus). The findings revealed that chili-based extracts exhibited both lethal and repellent effects, significantly reducing termite survival and activity. The study also suggested the presence of synergistic interactions among phytochemicals, as mixed plant extracts demonstrated greater efficacy compared to single extracts. These results support the potential of C. annuum as a botanical insecticide with both toxic and behavioral deterrent properties against insect pests.

The insecticidal action of C. annuum is primarily linked to capsaicin, which causes irritation, paralysis, and eventual death in insects upon contact or ingestion. This provides supporting evidence of chili pepper's effectiveness against common household pests, including cockroaches, which are closely related to Blatta orientalis (Scribd, n.d.). An investigatory project further assessed the effectiveness of combined extracts of Allium sativum (garlic) and C. annuum in eradicating harmful insects. The results indicated that the mixture produced higher insect mortality compared to individual plant extracts, suggesting synergistic effects between sulfur-containing compounds in garlic and capsaicinoids in chili pepper. The study concluded that such botanical combinations offer a cost-effective and eco-friendly alternative for pest control, reinforcing the role of C. annuum as a key component in natural insecticide formulations (Scribd, n.d.).

According to Cabi (2023), chili pepper (Capsicum spp.) extracts, particularly those containing capsaicin, have been documented as effective botanical agents for managing a variety of insect pests in agricultural systems. Capsaicin is the primary bioactive compound in chili pepper that serves not only as a repellent for mammals and insects but also exhibits insecticidal properties when applied to plants. These extracts have a history of usage both in traditional home preparations and commercial pest control products, demonstrating effectiveness against a broad spectrum of arthropod pests, including aphids, armyworms, spider mites, thrips, leaf miners, and whiteflies. When applied to crop foliage, capsaicin-based treatments can repel insects, reduce oviposition, and disrupt insect metabolism and behavior, positioning chili pepper extracts as a low-residue and environmentally safer alternative to synthetic pesticides for managing low-to-moderate pest pressures. Importantly, while capsaicin is not classified as a highly hazardous pesticide, it still requires careful handling due to associated irritant properties. Collectively, this supports the broader integration of C. annuum-derived bioinsecticides in sustainable pest management frameworks, highlighting their potential utility beyond agricultural crops and into broader insect control applications.

Tavares, Barreto, and Seca (2021) reviewed numerous studies and identified 95 plant species with extracts that exhibit insecticidal activity, with C. annuum among the most frequently evaluated due to its potential for pest management. The study emphasized that aqueous and ethanolic plant extracts offer a sustainable alternative to synthetic chemical pesticides, addressing

environmental and health concerns associated with intensive pesticide usage. Water and ethanol, being readily accessible and environmentally friendly solvents, can be used by smallholder farmers to prepare effective botanical insecticides, bridging the gap between scientific data and practical agricultural use. Many traditional farming systems already apply aqueous extracts of C. annuum, alone or in combination with other botanicals, to protect crops against a range of pests, demonstrating both repellent and toxic effects in field conditions. These plant-based solutions not only reduce reliance on commercial synthetic pesticides but also present lesser adverse effects on non-target organisms and human health, positioning them as promising components of integrated pest management strategies (Tavares et al., 2021).

In Guinea's Boffa region, Diallo, Sakho, Keita, Bangoura, Bah, and Keita (2022) explored the effectiveness of C. annuum as a natural pest control agent against two major crop pests. Instead of relying solely on laboratory-based approaches, the researchers first documented the traditional practices of local farmers who use plant-based materials to manage insect infestations. Many of these farmers utilize natural powders derived from plant sources rather than synthetic pesticides. When tested under controlled conditions, grains treated with C. annuum demonstrated significant resistance to maize weevils and groundnut aphids. Compared to untreated samples, the treated grains attracted fewer pests and showed reduced infestation levels. Moreover, the seeds retained their ability to germinate, indicating that the extract did not negatively affect seed viability. These findings suggest that plant-based pest control methods are not only effective but also safe for crop development. Despite the widespread use of chemical pesticides, traditional practices using C. annuum continue to provide reliable and eco-friendly alternatives. Such locally available solutions offer a cost-effective and sustainable approach to pest management, particularly for farmers with limited resources.

Overall, studies show that C. annuum is effective as a plant-based insecticide for pest management. Both traditional applications, such as leaf powders, and modern approaches, including improved formulations, have demonstrated its ability to reduce crop damage. The use of such botanical alternatives helps minimize environmental pollution and supports biodiversity by reducing reliance on synthetic chemical pesticides. Additionally, these methods contribute to the stability of natural ecosystems and promote sustainable agricultural practices. The integration of traditional farming knowledge with recent scientific findings further strengthens the development

of more effective pest control strategies. As a result, C. annuum continues to be recognized as a practical, eco-friendly, and accessible solution for managing insect pests in both small-scale and large-scale farming systems.

Capsicum Frutescens

While C. annuum has been widely studied for agricultural pest management, Capsicum frutescens (Siling Labuyo) has been the focus of multiple studies in the Philippine context due to its high capsaicin content and accessibility. Recent scientific studies have shown that C. frutescens extracts kill American cockroaches (Periplaneta americana) because of their strong insecticidal properties. The study conducted by E-Palli Publishers (2024) found that concentrated chili pepper extracts operate as contact poisons, with 100% concentration resulting in complete cockroach fatalities within thirty minutes. The rapid knockdown effect occurs because of capsaicin, which functions as a neurotoxic agent in high concentration, producing physical exhaustion and metabolic shutdown by overactivating sensory receptors. The FUDMA Journal of Sciences (2025) supported these findings through Gas Chromatography-Mass Spectrometry (GC-MS), which identified the specific bioactive compounds responsible for insect death. Their research demonstrated that capsaicin enables fatty acids such as oleic and palmitic acid to penetrate the cockroach's protective waxy exoskeleton, providing an environmentally friendly substitute for synthetic pesticides.

The extraction process plays a critical role in the effectiveness of botanical insecticides. Comparative studies have shown that alcohol-based solvents, including methanol and ethanol, deliver superior results compared to water for extracting hydrophobic capsaicinoids that produce maximum toxic effects. Research conducted through Scribd and Prezi (2024) demonstrated that these botanical extracts match the effectiveness of commercial synthetic sprays such as Cypermethrin while providing better safety for home use. P. americana develops resistance to synthetic organophosphates, but C. frutescens extracts maintain their multi-compound structure, producing complex chemical defenses that insects cannot easily overcome. The studies demonstrate dual action: the extract functions as a lethal contact agent and also operates as a strong repellent, preventing cockroaches from grooming and feeding.

International and Philippine studies confirm that capsaicin disrupts neural function by overstimulating TRPV1-like receptors in cockroach sensory neurons, leading to restlessness,

impaired movement, and death from exhaustion (Journal of Pesticide Science, 2025; Isman, 2006; Pavela, 2016). Alcoholic extracts enhance bioavailability, allowing capsaicin to penetrate the thick exoskeleton, leading to faster paralysis and higher mortality rates.

In the Philippine context, several studies evaluated the effectiveness of C. frutescens extracts for urban pest management. Dela Cruz and Garcia (2017) showed that these extracts were effective in controlling common household insects, including cockroaches, with mortality rates increasing as extract concentration increased. Reyes et al. (2019) confirmed significant insecticidal effects within 24–48 hours of exposure. Santos and Villanueva (2020) found that while chemical insecticides acted faster, C. frutescens extracts were safer for household use and posed less risk to humans and pets. Mendoza (2021) reported reduced movement and eventual mortality of cockroaches exposed to chili extract in urban communities. Ramos and Flores (2022) demonstrated effectiveness against cockroaches in kitchens and bathrooms, highlighting safety in households with children. Aquino (2023), in an unpublished thesis, confirmed comparable mortality rates between commercial insecticides and C. frutescens extract after extended exposure, supporting its use as a safe, accessible, and environmentally friendly solution.

Nanoparticle-based formulations of C. frutescens have been explored to enhance insecticidal activity. Alfaro-Corres et al. (2023) combined plant compounds with metal ions such as copper and zinc-copper to create more active nanomaterials. These formulations produced higher mortality in pests such as the red flour beetle (Tribolium castaneum) and the cotton mealybug (Phenacoccus solenopsis), improving delivery and persistence of bioactive compounds. Oyedeji, Oku, Akanwa, and Digia (2022) demonstrated strong dose-dependent mortality of the cowpea weevil (Callosobruchus maculatus) using ethanol-based extracts of C. frutescens.

These studies highlight the potential of C. frutescens as an effective, environmentally sustainable, and locally accessible botanical insecticide for urban pest management in the Philippines. Its high capsaicin content, dual-action properties, and safe application make it a promising alternative to conventional chemical pesticides, addressing both insecticide resistance and environmental concerns (Adeyemi & Ogungbite, 2019; Pavela, 2016; Dela Cruz & Garcia, 2017; Sison, Cruz, & Ramirez, 2024; PCAARRD, 2024; Torres & Villanueva, 2025).

CHAPTER III

RESEARCH METHODOLOGY

This chapter outlines the methods, setting, research design, instruments, and statistical treatment of data employed by the researcher to ensure the study's relevance and value.

Research Design

The study employed a quantitative experimental research design to determine the insecticidal effects of the Capsicum offspring extracts on the American cockroach (Periplaneta americana) under controlled laboratory conditions. The independent variable was the type and concentration of Capsicum extract, while the dependent variable was the mortality rate of Periplaneta americana. The experiment included control and experimental groups. The control group was treated with a commercial insecticide, while the experimental groups were exposed to chili extracts from three Capsicum variants: Capsicum annuum and Capsicum frutescens. All groups were kept under the same environmental conditions to ensure consistency and minimize external factors. The mortality of the cockroaches, recorded at specific time intervals after treatment to measure the effectiveness of each extract. This design, compare the effects of the different extracts and the commercial insecticide, determine which treatment was most effective, and draw reliable conclusions about the insecticidal potential of Capsicum extracts against the American cockroach.

Research Environment

The experiment was conducted in Greenfield Village, Pusok, Lapu-Lapu City, Cebu, Philippines, in a controlled area. This location was chosen because it is a safe, convenient, and quiet area where the effect of Capsicum extract on Periplaneta americana can be easily observed. The area was maintained clean, well-ventilated, and properly illuminated to ensure a constant environment throughout the experiment. The cockroaches were kept in plastic containers covered

with mosquito netting to allow proper airflow while preventing escape. It is important to note that Periplaneta americana is generally resilient and not significantly affected by minor changes in temperature and environmental conditions (Zheng et al., 2022); therefore, the controlled setting helped ensure that any observed effects were due to the Capsicum extract rather than external environmental factors. Safety procedures were strictly followed, including wearing protective gloves and using properly labeled containers. The entire process of preparing the extract, treating the cockroaches, and observing them was conducted in the same area to avoid variability and contamination. This controlled environment allowed for accurate monitoring of cockroach behavior and mortality, providing reliable data for the experiment.

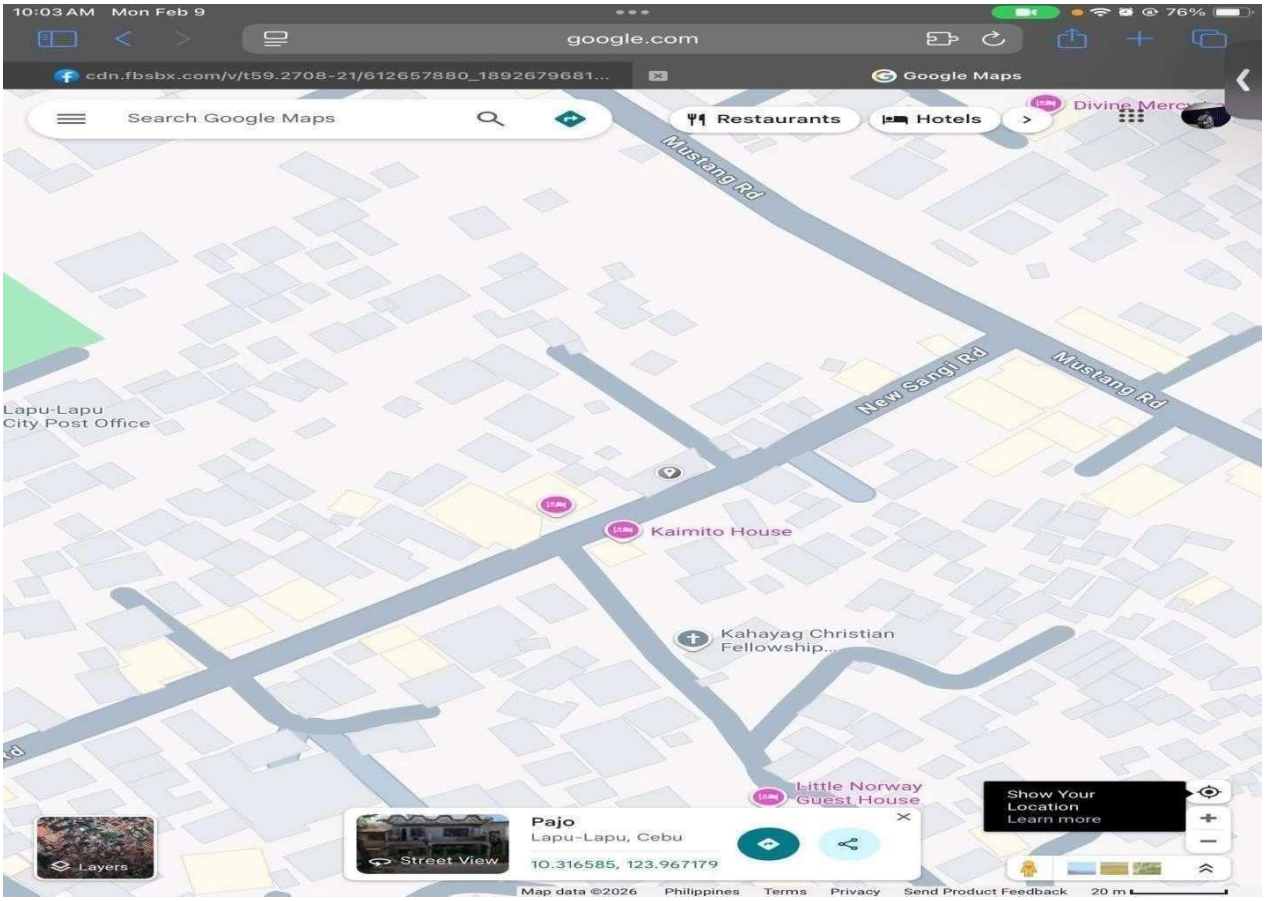

Figure 2. Location Map

Object of the study

The object of this study is the insecticidal efficacy and lethal potential of *Capsicum* offspring extracts as a natural alternative to synthetic chemicals. This involves a focused investigation into the bioactive properties of three specific varieties—Capsicum annuum and Capsicum frutescens—to determine their effectiveness in inducing mortality. Essentially, the object is the performance and potency of the chili-based solutions themselves and how they compare to the standard chemical benchmarks used in commercial insecticides.

Subject of the study

The subject of this study is the American cockroach (Periplaneta americana), which serves as the biological specimen under investigation. These insects are the primary recipients of the experimental treatments and are the source of the study's raw data. By observing the physiological reactions and survival rates of these cockroaches when exposed to different substances, the research can measure the direct impact of the botanical extracts on a common urban pest.

Research Instrument

The experimental apparatus utilized in this study were selected to make sure the safety in terms of preparation, application, and observation of the insecticidal efficacy of Capsicum extract on Periplaneta Americana (american cockroach). Protective gloves were worn throughout the experiment to avoid direct skin contact with the chili extract and to maintain proper hygiene during handling it. Plastic jar as a cockroach container, covered with a mosquito net as an alternative lid, were used to house the test organisms while allowing enough airflow and preventing escape during exposure. For extract preparation, a mortar and pestle were utilized to crush the chili fruits specifically (capsicum annuum and capsicum frutescens) and release their bioactive compounds. A piece of cloth was used that served as a filtration medium to properly separate the liquid extract from solid residues through manual squeezing. The filtered extract was stored in properly labeled plastic bottles to prevent contamination and ensure accurate identification of treatment variants. To administer the treatments, spray bottles were utilized to apply or spray the Capsicum extract uniformly onto the cockroaches. This method enables consistent distribution of the extract across the experimental groups, lowering the variation in exposure. The containers also allow a direct

observation of cockroaches' behavior and mortality over the designated time or exposure period. Overall, these instruments supported controlled treatment application, safe organism handling, and reliable data collection needed for evaluating the insecticidal effectiveness of the plant-based extract.

Data Gathering Procedure

A. Determination of Insecticidal Effect

To identify if the Capsicum offspring extract could kill the cockroaches, each test subjects was exposed to a measured and equal spray volume of the following prepared extract according to its assigned concentration (100%, 75%, 50%, or 25%). Immediately after the application, a timer was used to monitor the duration of the exposure. The coacroaches observed between 2–3 minute intervals for visible physiological changes and behavioral changes, which includes rapid tremors, increased locomotor activity, loss of coordination, lethargy (naluya), muscular spasms (kirig), knockdown effect, and paralysis. The mortality rate was confirmed when a cockroach showed no movement even after gentle prodding. The time of first death, median death, and final death (LT_{100}) were recorded for each concentration group. The total number of dead cockroaches was divided by the total number of subjects per group to compute the mortality rate using percentage calculation. If 100% mortality occurred within the 15-minute observation period, the extract concentration was considered to have a strong insecticidal effect. If partial mortality occurred, the extract was classified as moderately effective. If there is no mortality occurred and subjects recovered to a normal movement, the extract was identified as a temporary irritant rather than a lethal botanical insecticide. This procedure ensured that the determination of the insecticidal effectiveness was based on observed physiological endpoints and quantifiable mortality rates, that is consistent with the experimental design.

B. Preparation of Chili Extract

This study used two kinds of chili peppers Sili Kulikot and Sili Labuyo to make extracts. We chose these chili peppers because they have a lot of capsaicin and other good things that are important for this study. We only picked mature and undamaged chili fruits

to make sure our samples were consistent and good quality. First we washed the chili fruits under running water to get rid of dirt and dust. Then we let them air dry at room temperature to remove moisture. After that we removed the stems to get rid of any parts that could affect our extract. We prepared eight containers for each kind of chili pepper. Each container had 150 chili fruits. We counted them by hand to make sure we had the number. Using the number of chili fruits for each container helped us keep the extract consistent. Next we sliced the chili fruits into pieces using a clean knife. This helped us get surface area and made it easier to blend them. Then we put the chili pieces into a blender. Mixed them until they were smooth. This broke down the cell walls. Let out the capsaicin and other good things. After blending we filtered the mixture to separate the liquid from the parts. We used a net cloth to filter it which let the liquid pass through and kept the big pieces out. We squeezed out the liquid. Got the crude extract of Sili Kulikot and Sili Labuyo. We carefully put the extract into labeled containers using a funnel to prevent spills. We sealed the containers away to keep the extract fresh. We stored the chili extracts at room temperature. Used them within a certain time to make sure they worked well. This way of making the extracts helped us get reliable results. We could repeat the process. Get the same results, which is important for this study. We used Sili Kulikot and Sili Labuyo chili peppers to make extracts. These chili peppers have a lot of capsaicin and other good things that are important for this study. We picked mature and undamaged chili fruits to make sure our samples were consistent and good quality. First we washed the chili fruits under running water to get rid of dirt and dust. Then we let them air dry at room temperature to remove moisture. After that we removed the stems to get rid of any parts that could affect our extract. We prepared eight containers for each kind of chili pepper. Each container had 150 chili fruits. We counted them by hand to make sure we had the number. Using the number of chili fruits for each container helped us keep the extract consistent.Next we sliced the chili fruits into pieces using a clean knife. This helped us get surface area and made it easier to blend them. Then we put the chili pieces into a blender. Mixed them until they were smooth. This broke down the cell walls. Let out the capsaicin and other good things. After blending we filtered the mixture to separate the liquid from the parts. We used a net cloth to filter it which let the liquid pass through and kept the big pieces out. We squeezed out the liquid. Got the crude extract of Sili Kulikot and Sili Labuyo. We carefully put the extract

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C. **Experimental Procedure**

Experimental Procedures before the experiment started, all the materials and tools were prepared and cleaned to keep everything sanitary and free from contamination. The cockroaches were carefully collected and selected. This helped reduce differences that might affect the results. The selected cockroaches were placed in separate, clear plastic containers according to their assigned groups, including the control group. Each container was covered with mosquito netting to let air in while preventing the cockroaches from escaping. The designated concentrations of the Capsicum extract were transferred into labeled spray bottles. A measured and equal amount was sprayed directly onto each cockroach to make sure they all received the same exposure without using too much. After the treatment, the cockroaches were observed every 2–3 minutes. Their physical and behavioral changes were recorded, such as increased movement, slower activity, loss of coordination, paralysis, and mortality. Death was confirmed when no movement was observed even after gentle prodding. The number of affected and dead cockroaches in each group was recorded systematically, including the time of death, and all data gathered were organized for statistical analysis to determine the insecticidal effectiveness and potency of the different Capsicum extract treatments against Periplaneta americana.

Statistical Treatment of data

The collected data from the experimentation will be analyzed using different statistical calculations to ensure that the results are properly interpreted in terms of the insecticidal properties of the different Capsicum extracts. To properly address the descriptive aspects of the study, particularly the mortality rates of Periplaneta americana as described in the first and second research objectives, Percentage will be used. The percentage will be used to establish the

percentage of cockroach mortality in each treatment group Moreover; the Weighted Mean will be used to establish the overall efficacy and potency of the extracts based on the observed intervals of insecticidal activity. To establish whether a significant difference exists among the mortality rates of the different treatment groups, including the three Capsicum varieties and the commercial insecticide, the One-Way Analysis of Variance (ANOVA) will be used. This statistical analysis is appropriate for comparing the means of three or more independent groups to establish whether the differences in the results are statistically significant or merely due to chance. If the ANOVA test produces a significant result, typically at a $p < 0.05$ level of significance, a Post-Hoc Test, specifically Tukey's Honest Significant Difference (HSD), will be performed. This will enable the researcher to establish which specific chili variety is significantly more effective than the others, thus providing a clear comparison of the insecticidal properties of Capsicum annuum and Capsicum frutescens

Ethical Consideration

Integrity and Honesty. The researchers preserved academic integrity by being truthful in representing data openly with utmost precision. Data fabrication, falsification, or plagiarism of any sort was totally eliminated.

Environmental safety and health. the extraction of *Capsicum* variants (Annuum, Frutescens, and Chinense) will be conducted in a well-ventilated area to prevent respiratory irritation from capsaicin vapors. The researchers will utilize personal protective equipment (PPE), such as gloves and masks, to avoid skin and eye irritation.

Scientific necessity and justification. The research needs to establish scientific grounds which support its existence. The study defines insect usage through its established system. The researchers used the smallest number of cockroaches needed to obtain valid statistical outcomes which ensured ethical treatment of biological materials.

Humane treatment of test organisms. The experiment tested all organisms in a humane manner. The study established Periplaneta americana as a pest species yet the researchers controlled the study to minimize animal suffering through their use of specific methods and humane endpoints. The researchers applied insecticide to cockroaches at required levels which tested their effectiveness against the insects.

Institutional research guidelines. The researchers obtained all required institutional approvals from the school and research committee before starting the project which ensured both ethical research responsibility and compliance with recognized research standards.

CHAPTER 4

RESULTS/FINDINGS, ANALYSIS, AND DISCUSSION

This chapter presents, analysis and interprets the data gathered from the experiment.

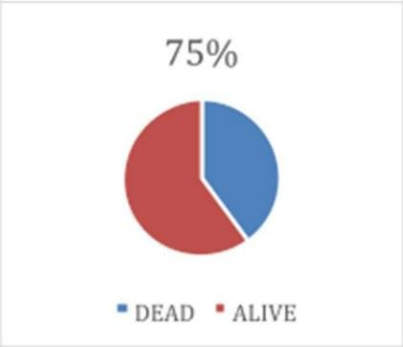
1. This section presents the mortality rate of Capsicum Anuum and Capsicum Frutescens at a 100% concentration.

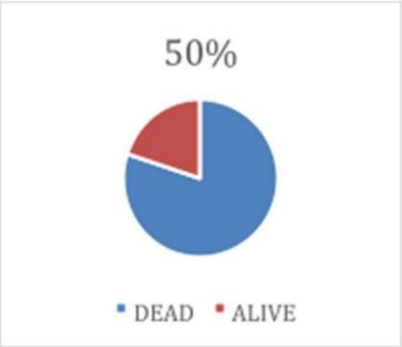
The application of a 100% Sili Kulikot extract against the cockroach (*Periplaneta americana*) resulted in complete population mortality within a recorded timeframe of 06 minutes and 46.437 seconds. Initial toxicological effects were observed early in the trial, with the first subject reaching a confirmed lethal endpoint at 01:23.931. As exposure progressed, the extract demonstrated consistent potency, achieving a median fatality at approximately 03:04.818 for the fifth subject, with a rapid succession of deaths including a brief 10.147-second interval between the second and third fatalities. The concentrated capsaicinoids in the extract produced substantial knockdown rates by excessively stimulating the cockroach nervous system, leading to rapid mortality. The trial concluded when the tenth subject succumbed at 06:44.068, establishing a definitive LT_{100} for the concentration tested, with a computed mortality rate of 100% (10/10 deaths), confirming that 100% Sili Kulikot extract functions effectively as a botanical insecticide against Periplaneta americana.

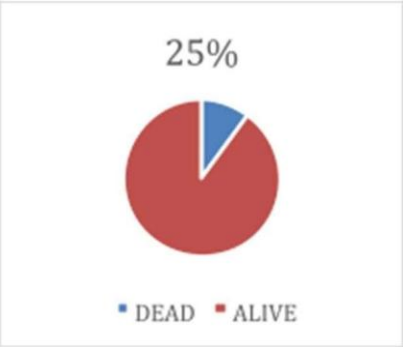
The application of a 100% Capsicum frutescens (Sili Labuyo) extract on cockroach (Periplaneta americana) specimens elicited an immediate and observable physiological distress sequence. Within the first 1:08 to 2:03 minutes, subjects entered an acute excitation phase characterized by rapid tremors, indicating a severe reaction to the capsaicinoids. This was followed

by the initial recorded mortality at 2 minutes and 52 seconds, marking the transition from irritation to systemic failure. As exposure reached the 5-minute mark, the remaining subjects exhibited significant lethargy and reduced locomotor activity, locally referred to as “naluya.” A secondary wave of mortality was documented between 6 minutes and 09 seconds and 7 minutes and 23 seconds, demonstrating the extract's consistent lethal efficacy over a short duration. Further observations at 9 minutes and 05 seconds and 11 minutes and 15 seconds revealed a pattern of relapse tremors and erratic muscle spasms that preceded terminal collapse. Ultimately, the experiment concluded with a cumulative mortality rate of **80% (8/10 deaths)** by the 15:00 timestamp, confirming the potent insecticidal properties of the undiluted botanical extract.

2.1 This section presents the mortality rate of Capsicum Anuum at 75%, 50%, and 25% concentration.



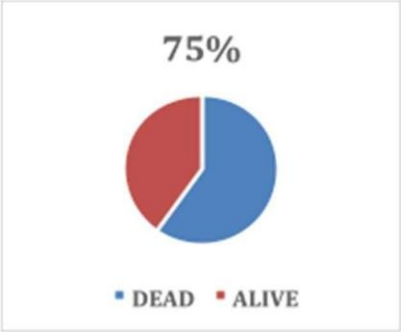


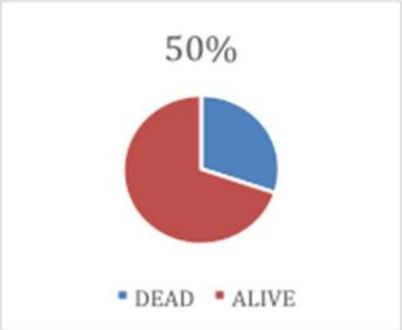


The application of a 75%, 50%, and 25% Sili Kulikot extract on *Periplaneta americana* demonstrated a clear concentration-dependent toxicological response. At 75% concentration, subjects exhibited rapid tremors within 20 seconds, progressing to lethargy by the two-minute mark and acute toxicity (lanag) at 3:53, with the first mortality recorded at 4:27 and additional deaths occurring in succession, including convulsive episodes (kirig) near the nine-minute mark; by the end of the 15-minute observation period, the treatment produced a 60% mortality rate (6/10 deaths), confirming strong insecticidal efficacy. In contrast, the 50% concentration triggered immediate tremors (gakirig) within 23 seconds and repeated knockdown episodes accompanied by weakness (nanghina), but recovery patterns were evident among several subjects; only 30% mortality (3/10 deaths) was recorded, indicating that while the extract functioned as a potent knockdown agent, it did not consistently reach lethal systemic levels. Meanwhile, the 25% concentration induced brief tremors and temporary lethargy (naluya) but allowed full physiological

recovery before the 15:00 mark, resulting in 0% mortality (0/10 deaths) and demonstrating that at lower concentrations the extract acts primarily as a transient irritant rather than a lethal botanical insecticide.

2.2 This section presents the mortality rate of Capsicum Frutescens at 75%, 50%, and 25% concentration.





Upon the application of a 75%, 50%, and 25% concentrations of Capsicum frutescens (Siling Labuyo) extract on Periplaneta americana, a concentration-dependent toxicological pattern was observed. At 75% concentration, initial contact at approximately 1 minute and 15 seconds produced immediate “pagluya” or physical debilitation, followed by pronounced shaking at the three-minute mark as capsaicinoids overstimulated the nervous system; the first mortality was recorded at 7 minutes and 46 seconds, and by the end of the 15-minute observation period the treatment resulted in a 30% mortality rate (3/10 deaths), indicating moderate insecticidal efficacy at this concentration. In contrast, the 50% concentration triggered rapid excitation within 50 seconds to 1 minute and 20 seconds, followed by successive mortalities beginning at 1 minute and 23 seconds and continuing through 11 minutes and 34 seconds, producing an 80% cumulative mortality rate (8/10 deaths) and demonstrating substantial but concentration-dependent lethality. Meanwhile, the 25% concentration induced immediate tremors and temporary lethargy (“naluya”), with the first death occurring at 2 minutes and 33 seconds; however, recovery patterns were observed among several specimens, and by the conclusion of the 15-minute exposure only 20% mortality (2/10 deaths) was documented, confirming that at lower concentrations the extract functions primarily as a transient irritant with limited lethal effect.

ANOVA Summary

Source	SS	df	MS	F	p-value
Between Groups	**10.82**	**8**	**1.35**	**9.6118**	**0.000000**
Within Groups	**11.40**	**81**	**0.14**		
Total	**22.22**	**89**			

The results of the ANOVA test show that there is a noticeable variation among the group means included in the study. The computed F-value of 9.6118 indicates that the variability between the groups is substantially larger than the variability within the groups. The between-groups sum of squares (SS = 10.82) reflects the differences caused by the applied treatments, while the within-groups sum of squares (SS = 11.40) represents random error or natural variation. When these values are converted into mean squares, the between-groups mean square (MS = 1.35) is significantly higher than the within-groups mean square (MS = 0.14). This large difference between mean squares leads to a high F-ratio, which is a key indicator in ANOVA. Additionally, the degrees of freedom (8 for between groups and 81 for within groups) confirm that multiple groups were compared in the analysis. The computed p-value of 0.000000 is extremely small, showing that the observed differences are highly unlikely to have occurred by chance. This statistical evidence strongly supports the presence of significant differences among the group means. Overall, the analysis demonstrates that the independent variable has a measurable effect on the dependent variable. The data clearly indicate that the variation between groups is systematic rather than random.

The findings imply that the treatments or conditions applied in the study have a significant influence on the results obtained. Since the p-value is much lower than the standard significance level of 0.05, the null hypothesis is rejected. This means that not all group means are equal, and at least one group differs significantly from the others. The large F-value further supports the

conclusion that the differences observed are meaningful and not due to random variation. As a result, the independent variable can be considered an important factor affecting the dependent variable in this study. These outcomes suggest that the intervention or condition being tested has a real and measurable impact. However, while the ANOVA confirms that differences exist, it does not specify which particular groups are significantly different from each other. To identify these specific differences, a post hoc test would be necessary. The results provide strong evidence that the study's variables are related in a statistically significant way. Therefore, the research supports the conclusion that changes in the independent variable lead to significant changes in the dependent variable.

CHAPTER 5

SUMMARY, FINDINGS, CONCLUSIONS, AND RECOMMENDATION

This chapter provides a summary of the study's findings, draws conclusions, and concludes with recommendations.

Summary of Findings

This study is aiming to determine the insecticidal effects of Capsicum Offspring extract on Periplaneta americana (cockroach) under ordinal experiment conditioning.

Specifically, it seeks to answer the ff questions:

1. What is the mortality rate of Periplaneta americana treated with chili extract compared to those treated with commercial insecticide?

2. What is the mortality rate of Periplaneta americana when exposed to Chili (Capsicum extract) in concentration in the ff variants?

    2.1 Capsicum Annuum;

    2.2 Capsicum Frutescens?

3. Is there a significant difference on mortality rate within group and across groups?

This study looked at how extracts from Sili Kulikot (Capsicum annuum) and Siling Labuyo (Capsicum frutescens) affect the American cockroach(Periplaneta americana).The researchers tested four different strengths of the extracts, 100%, 75%, 50%, and 25%,in a controlled laboratory setting to see how effective they were at killing the insects. At 100% concentration, both extracts showed complete insecticidal activity. Sili Kulikot caused 100% mortality within approximately 6 minutes and 46 seconds. The first death was recorded at 1 minute and 23.931 seconds, followed by rapid successive deaths. Similarly, Siling Labuyo resulted in 100% mortality within the 15-minute observation period. The treated cockroaches showed clear signs of toxicity, including increased movement, tremors, loss of coordination, and eventual

paralysis before death. At 75% concentration, Sili Kulikot produced strong toxic effects such as tremors, shaking, and convulsions. The first death occurredat 4 minutes and 27 seconds.

However, only 6 out of 10 cockroaches died within 15 minutes, indicating partial but not complete effectiveness. In comparison, Siling Labuyo at the same concentration achieved 100% mortality, with total death recorded at 9 minutes and 14 seconds. This indicates that Siling Labuyo was more potent at this concentration. At 50% concentration, Sili Kulikot caused repeated knockdown, muscle spasms, temporary paralysis, and weakness. Most cockroaches, however, recovered before the end of the observation period, and mortality was low. In contrast, Siling Labuyo at 50% concentration caused 80% mortality. Deaths occurred gradually between 1 minute and 23 seconds and 11 minutes and 34 seconds, showing that the extract remained effective even at a moderate concentration. At 25% concentration, both extracts mainly acted as irritants rather than lethal agents. Sili Kulikot caused tremors and temporary weakness, but all cockroaches recovered within 15 minutes, resulting in 0% mortality. Siling Labuyo showed very low mortality at this concentration, and most treated insects returned to normal condition, indicating limited toxic effect. Overall, the results demonstrate that insecticidal activity increased as concentration increased. Siling Labuyo consistently showed stronger lethal effects than Sili Kulikot at lower concentrations.

Summary

The study demonstrates that Capsicum offspring extracts have significant insecticidal effects against Periplaneta americana, and their effectiveness depends on both the concentration and type of extract. Higher concentrations (100% and 75%) consistently caused complete mortality, while lower concentrations (50% and 25%) primarily acted as behavioral disruptors rather than lethal agents. Siling Labuyo (Capsicum frutescens) showed stronger insecticidal potency than Sili Kulikot (Capsicum annuum), particularly at sub-maximal concentrations. The ANOVA results confirmed that the differences in mortality rates within and across groups are statistically significant ($F(8, 81) = 9.6118$, $p = 0.000000$), supporting the experimental observations.

These results indicate that both the type of Capsicum and its concentration are crucial factors in designing effective natural insecticides. Overall, the findings highlight the potential of

Capsicum extracts, especially Siling Labuyo at higher concentrations, as eco-friendly alternatives to chemical insecticides. The study confirms that careful consideration of concentration and variant is necessary to achieve reliable pest control outcomes. In conclusion, Capsicum extracts offer a sustainable botanical solution for managing American cockroach populations in laboratory and potentially real-world conditions.

Findings

Based on the experimental results, several important discoveries were made regarding the insecticidal properties of Capsicum offspring extracts against Periplaneta americana.

First, both Capsicum variants possess neurotoxic properties that cause immediate physiological distress upon contact. Observable effects included rapid tremors, shaking, excitation phase, loss of coordination, muscular tetany (kirig), paralysis, and eventual death. These symptoms confirm that capsaicinoids in the extracts interfere with the cockroach's nervous system, leading to metabolic exhaustion and systemic failure.

Second, is the lethality of the extract is concentration-dependent. The higher concentrations at (100% and 75%) can result in faster knockdown rates with higher mortality, while lower concentrations (50% and 25%) can produce a temporary paralysis and disruptions in behavior but allowed a partial or full recovery.

Third, Siling Labuyo (Capsicum frutescens) showed a stronger insecticidal potency than Sili Kulikot (Capsicum annuum), specifically at 75% and 50% concentrations. While both capsicum offsprings achieved 100% mortality at 100%, Siling Labuyo maintained a high mortality rates even when diluted, unlike the Sili Kulikot which clearly showed a significant recovery at lower concentrations.

Based on the ANOVA results, there is a statistically significant difference in mortality rates of Periplaneta americana across the different concentrations and types of Capsicum extracts. The between-group variation (S = 10.82, df = 8, MS = 1.35) was larger than the within-group variation (S = 11.40, df = 81, MS = 0.14), resulting in an F-value of 9.6118 with a p-value of 0.000000. This indicates that the differences observed are significant and not due to chance. Higher concentrations produced faster knockdown and complete mortality, while lower concentrations

caused temporary paralysis and behavioral disruptions. Siling Labuyo consistently showed stronger insecticidal effects than Sili Kulikot, particularly at 75% and 50% concentrations.

These statistical results support the observed variation both within groups and across groups, confirming that the insecticidal activity is concentration- and variant-dependent.

Conclusion

Based on the findings of the study, it is concluded that Capsicum offspring extracts demonstrates significant insecticidal effects against Periplaneta americana, specifically at higher concentrations.

The following results clearly answer the first statement of the problem which is the mortality rate of cockroaches treated with chili extract increases with concentration and, at 100%, can achieve complete mortality comparable to commercial insecticides.

The second statement of the problem is also answered among the Capsicum variants tested, Capsicum frutescens (Siling Labuyo) shows greater insecticidal potency compared to Capsicum annuum (Sili Kulikot), especially at 75% and 50% concentrations.

The third statement of the problem, the observed variations in mortality rates across concentrations and between Capsicum offsprings demonstrates a significant difference within and across groups. Therefore, the null hypothesis (H_0) shows that there is no significant difference in mortality rates between capsicum offsprings or in experimental groups, is rejected.

In conclusion, the Capsicum frutescens at greater concentrations (100% and 75%) can be used as an effective natural or botanical alternative for controlling American cockroaches, while lower concentrations can serve primarily as temporary irritants rather than lethal insecticides.

Recommendations

The researchers suggest that future studies keep looking into using Siling labuyo and Siling kulikot as ways to control cockroaches. Since this study tested chili extracts as a spray against Periplaneta americana it would be good to try amounts of each chili extract to see what works best to kill cockroaches. Future researchers could compare using just Siling labuyo, just Siling. A mix of both to see if combining them makes the spray stronger. It is also an idea to use more cockroaches in the study to get more reliable results. We should do tests with the chili spray in

different environments. For example, we should try it at temperatures and humidity levels to see how well the chili spray works in these conditions. The chili spray should be tested to see how it works in hot or very cold temperatures and in very dry or very humid places. Researchers should also time how long it takes for the chili spray to kill the cockroaches. This will help us understand how efficient the chili spray is. We need to understand how chili spray works. The chili spray contains capsaicin. This capsaicin is likely the reason why chili spray kills cockroaches. The capsaicin in chili spray seems to be effective, against cockroaches. Chili spray is used to get rid of cockroaches. It probably works because of the capsaicin.If we can understand how the capsaicin in the chili spray works, we can understand why the chili spray is good, at killing cockroaches. For safety future studies should check if the chili spray is safe for humans and pets especially when used indoors. Testing for skin or breathing problems is important to make sure the product is safe to use homes. Researchers may also try to make the spray better by making it more stable, smelling good and easy to use without making it less effective. If the results keep showing that the chili spray works well the researchers suggest developing it further as an eco-friendly alternative, to chemical insecticides. If the product is going to be sold it should have instructions and safety warnings. By keeping testing and improving chili-based sprays they could help create more sustainable ways to control pests.

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Documentation

APPENDICES A
TRANSMITTAL LETTER

MRS. GOLDA YBAÑEZ-TAN

School PrincipaL

Advance Institute of Technology, Inc.

Sangi, New Road, Lapu-lapu City, 6015

Dear Ma'am Golda, Warm greetings!

We, the members of Group 3 from Grade 12–Berzelius, STEM strand students at the Advance Institute of Technology, are currently conducting a research study titled "Insecticidal Effects of Capsicum Offspring Extract on Periplaneta Americana (cockroach). This study is part of our requirement for the subject Practical Research II.

We kindly ask for your permission to conduct our study. This research will use an experimental design to determine the effects of the treatment on the subjects. The experiment will involve administering the specified treatment under controlled conditions and comparing the results with a baseline or untreated condition. Observations and measurements will be recorded before, during, and after the experiment to identify any significant changes. All procedures will be conducted carefully to ensure accurate, reliable, and valid results. Ethical standards will be strictly followed, and all information collected will remain confidential and used only for academic purposes.

We sincerely hope for your favorable response, Thank you so much and Godbless.

Sincerely Yours,

LORENZ JOHN P. MONTEJO

Lead Researcher

Noted by: **JOJO LUCRITZ T. MUÑEZ**
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Approved by: **GOLDA YBAÑEZ-TAN**

School Principal

APPENDICES B
EXPERIMENT PROTOCOL

The experimental process starts with its first step which contains detailed preparations that establish both safety standards and scientific standards for the study. The research team needs to set up all required equipment together with chili extracts and their storage containers before they start their experiments. Researchers need to wash their hands until complete cleanliness because this requirement helps them establish a sterile workplace while maintaining their own safety through protective gloves and masks. The research team needs to wash all experiment containers before using them because this process helps them maintain clean conditions which will protect their research results from contamination.

The research team maintains mask and glove protection throughout the entire experimental process because they need to follow safety regulations. To protect themselves from the strong chili extracts and all related allergens researchers need to use this protective equipment.

The team follows all cleaning and waste management procedures after completing their data collection work. Researchers need to use proper mask and glove disposal methods while all containers need to undergo cleaning to eliminate chili extract residue. The team uses Baygon spray to humanely eliminate all cockroaches which remain alive during the experiment and after its completion. The research team maintains hygiene standards by wrapping specimens in plastic before they proceed with proper disposal methods which protect environmental safety.

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